

FURTON RESEARCH

Simulating a Committee of Expert Investors as an Investment Strategy

A Methodology Paper · Pre-Results Disclosure

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Abstract

Furton Research is a public, research-driven micro-investment project designed to transparently assess the viability of simulating the decision-making of a committee of expert investors as an investment strategy. The committee comprises five distinct investors, each simulated by Anthropic’s Claude grounded exclusively in primary sources authored by the individuals or their funds. This paper documents the system as built: the construction of five primary-source libraries, a four-phase committee engine that produces independent verdicts before structured deliberation, a retrieval-augmented generation (RAG) pipeline that grounds each agent in the most relevant passages for a given security, and a disciplined process for deploying and rebalancing a real \$10,000 portfolio benchmarked against the Dow Jones Industrial Average. We report the engineering decisions, the economic trade-offs of operating large language models at scale—including a detailed account of the tension between retrieval and prompt caching—and the full methodology by which results will be generated. No performance findings are reported yet; this document establishes the apparatus and the pre-registered process so that subsequent results can be evaluated against a transparent, fixed methodology.

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1. Introduction and Research Question

Professional investment management is, at its core, a structured disagreement among people who see the same facts through different frameworks. A value investor and a disruptive-growth investor can examine the same company and reach opposite conclusions, each internally coherent. Furton Research asks a specific, testable question: if the distinctive reasoning of several well-documented expert investors can be simulated by a large language model grounded in those investors' own writing, does a committee of such simulated investors produce investment decisions worth acting on?

The project is deliberately public and transparent. Rather than asserting that an artificial committee can or cannot invest well, it builds the apparatus, commits to a fixed process in advance, deploys real capital, and reports what happens against a clear benchmark. This paper documents everything up to the point of results: what was built, why each design choice was made, and the exact process by which a \$10,000 portfolio is managed and compared to the Dow Jones Industrial Average.

The central claim under examination is narrow and falsifiable: *that a committee of five primary-source-grounded simulated investors, deliberating over Dow 30 constituents, generates a portfolio whose risk-adjusted performance can be measured against the index it draws from.* Whether that claim holds is the subject of future reporting. This document concerns only the method.

1.1 What this project is not

Three disclaimers frame the work. First, this is a research project, not investment advice; the committee's outputs are simulations of documented reasoning styles, not vetted recommendations. Second, the simulated investors are not the real individuals and do not represent their current views; they are language-model personas constrained to primary-source text. Third, the committee's composition introduces a structural bias that the project treats as a finding to be studied rather than a flaw to be hidden.

2. The Committee and the Primary-Source Principle

The committee comprises five investors chosen to span complementary, and frequently conflicting, frameworks. Each is simulated by Anthropic's Claude, constrained to reason from a library of that investor's own primary-source material. The deliberate design goal is genuine disagreement: a panel that always agreed would carry no more information than a single agent.

Investor	Framework	Primary-source corpus
Warren Buffett	Quality value, durable moats, owner earnings	Berkshire Hathaway shareholder letters (1977–2023)
Howard Marks	Market cycles, risk, second-level thinking	Oaktree memos and published writing
Joel Greenblatt	Quantitative value, the “magic formula”	Three books on special-situation and value investing

Investor	Framework	Primary-source corpus
Cathie Wood	Disruptive innovation, exponential cost curves	ARK Big Ideas reports and interview transcripts
Leopold Aschenbrenner	AI infrastructure, regime shifts, scaling	“Situational Awareness” and a long-form interview

The primary-source principle is the project’s methodological backbone. Each simulated investor reasons only from text the investor or their fund actually produced. This is what separates the project from prompting a model to “act like Warren Buffett,” which would draw on the model’s diffuse training-data impression of the persona. By grounding each agent in a curated corpus and instructing it to reason from documented positions, the simulation is anchored to evidence rather than caricature.

2.1 Library construction

Each corpus was assembled by collecting primary-source documents, segmenting them into passages (“chunks”) of roughly 600–1,000 words, and storing each chunk with a citation and metadata in a structured manifest. The five libraries vary in size according to the available material:

Investor	Chunks (filtered)	Source character
Warren Buffett	90	High-doctrine filter applied to 405 total chunks
Leopold Aschenbrenner	108	Single treatise plus interview, no filter
Howard Marks	157	High-doctrine filter applied to 292 total chunks
Joel Greenblatt	166	Three complete books
Cathie Wood	165	Hybrid: ARK reports plus interview transcripts

For two investors a “high-doctrine” filter was applied, retaining only chunks expressing durable investment philosophy and discarding transactional or dated passages. The Cathie Wood corpus required a hybrid approach: because ARK’s flagship reports are graphically dense and text-thin, interview transcripts were added to capture her reasoning in prose. These construction choices are documented because they materially shape what each agent can retrieve and therefore how it reasons.

2.2 A note on the committee’s composition bias

The five frameworks cluster around company-level analysis—quality, value, innovation—with a single voice (Aschenbrenner) addressing structural technology shifts and only partial coverage of top-down macro conditions through Marks’s cycle lens. No member reasons primarily from interest rates, liquidity, or the broad economic cycle. This is a known limitation. It means the committee is expected to favor durable consumer and platform businesses and to under-weight deep-cyclical, commodity, and rate-driven names. Rather than correct this by adding members arbitrarily, the project treats the composition as fixed and its tilt as a measurable characteristic of the strategy.

3. The Committee Engine

A single evaluation proceeds through four phases. The design separates independent judgment from social influence, so that the committee genuinely deliberates rather than merely averaging five opinions.

3.1 Four phases

1. **Enrichment.** A fast model (Claude Haiku) gathers current market data on the security via web search and produces a structured investment brief—valuation, recent financials, competitive position, and material news. This step requires no primary-source grounding and is deliberately inexpensive.
2. **Blind evaluation.** Each active investor receives the brief and their retrieved primary-source context, then issues an independent verdict—a position (Buy, Pass, Short, or Abstain) and a conviction score from 1 to 10—without seeing any colleague’s view. This preserves the independence of each framework.
3. **Deliberation.** Each investor is then shown the others’ verdicts and theses and may respond to the most significant disagreement, revising conviction if warranted. This is the step that makes the panel a committee rather than a poll: members react to one another.
4. **Synthesis.** A high-capability model (Claude Opus) acts as committee secretary, writing a single official statement that records the final position, the argument that carried the majority, the most significant dissent, and the recommended action.

3.2 The weighted vote

The committee’s position is computed from member verdicts by a conviction-weighted vote. A Buy contributes positively in proportion to its conviction, a Short negatively, and Pass or Abstain contribute zero. The sum is normalized to a score between -1 and $+1$. Scores at or above $+0.3$ are read as a committee Buy; at or below -0.3 , a Short; in between, a Pass. The same threshold governs position sizing in the portfolio: a stronger committee score earns a larger allocation.

3.3 The two-stage filter

Running the full four-phase pipeline on every security is wasteful when the committee is plainly indifferent. A two-stage filter addresses this. After the blind vote, a stock advances to the expensive deliberation and synthesis phases only if it shows a real signal: a committee score at or beyond ± 0.3 , a genuine split (at least one Buy and one Short), or any single member at conviction 8 or higher. Securities that draw a uniformly low-conviction Pass are recorded with a deterministic note—“deliberation round skipped due to all-around low conviction”—and incur no further cost. This roughly halves the cost of a full market screen while concentrating effort on the decisions that matter.

4. Retrieval-Augmented Grounding

The most consequential technical decision in the project concerns how each agent receives its primary-source material. The naive approach injects a fixed slice of each investor’s corpus into every evaluation. This is both wasteful—most of the corpus is irrelevant to any given security—and limiting, because a fixed slice cannot adapt to the company under review.

4.1 How retrieval works here

The project uses retrieval-augmented generation (RAG). Every chunk in every library is converted to a numerical vector (an “embedding”) using a compact local model, all-MiniLM-L6-v2, that runs offline on commodity hardware with no per-use cost. At evaluation time, the investment brief is embedded and compared against the investor’s chunk vectors by cosine similarity; the ten most relevant chunks are retrieved and injected as that agent’s context for that specific security.

The effect is that when the committee evaluates a beverage company, the Buffett agent retrieves his writing on consumer brands and pricing power; when it evaluates a semiconductor firm, it retrieves his passages on circle of competence and technological change. Each agent reasons from the most relevant fraction of its corpus rather than a representative sample, and the full library—not a pre-truncated slice—becomes accessible.

A fallback preserves robustness: if embeddings have not been generated, the system reverts to evenly sampled chunks, tuned to return the same number of passages so that cost and behavior remain consistent. The system reports at startup which mode is active for each investor.

5. The Retrieval–Caching Trade-off and Token Economics

Operating five grounded language-model agents over deliberation rounds is the dominant cost of the project, and one trade-off proved central enough to warrant detailed treatment: retrieval and prompt caching work against each other.

5.1 Why the two conflict

Prompt caching is a provider feature that stores a block of input the first time it is sent and serves it cheaply on subsequent identical requests, dramatically reducing the cost of resending a large, unchanging context. It pays off only when the cached block is byte-for-byte identical between calls.

Retrieval guarantees the opposite. Because RAG retrieves different chunks for every security, each agent’s corpus block changes from one stock to the next. The cached block is never reused, so caching never produces a read hit—and worse, the act of marking a block for caching adds a one-time write premium of roughly 1.25× the input price. Under retrieval, that premium is paid on every call and never recouped: the worst possible caching outcome.

5.2 How the conflict surfaced

The conflict was not theoretical. An early full-market screen consumed roughly \$12.70 against an expectation near \$5. Inspection of the run log showed the cause unambiguously: every single call reported zero cache reads and a full cache write of around 70,000 tokens. Each security paid the write

premium and recouped nothing, because retrieval had made every context block unique. The original cost estimate had implicitly assumed warm-cache reads across the screen—an assumption retrieval silently invalidated.

5.3 The resolution

The fix makes caching conditional. When an agent uses retrieval—its context differing every call—the system sends the context as a plain block at the standard input rate and omits the cache marker entirely, eliminating the unrecoverable premium. When an agent falls back to the fixed even-sampling mode, where the context is stable across calls, caching is re-enabled and pays off as intended. The system inspects each investor’s retrieval status per call and chooses accordingly.

This change alone reduced input cost on the corpus by approximately twenty percent with no effect on output quality, by removing a premium that retrieval guaranteed would be wasted.

5.4 Per-stock and per-run economics

With retrieval active and caching correctly disabled on the retrieval path, the realized cost structure for a single security is approximately:

Phase	Model	Approx. cost	Share
Enrichment	Haiku	\$0.015	~5%
Blind vote (5 agents)	Sonnet	\$0.12–\$0.15	~45%
Deliberation (advancers)	Sonnet	\$0.11	~40%
Synthesis	Opus	\$0.015	~6%

The committee votes—ten grounded agent calls per security across blind and deliberation rounds—dominate cost; enrichment and synthesis are minor. This is why retrieval depth, not enrichment, is the lever that matters. At a retrieval depth of ten chunks per investor, a full real-time screen of the thirty Dow constituents costs on the order of \$9–\$10, with the two-stage filter sparing the many low-conviction names from the expensive second stage.

5.5 Batch processing as a cost lever

For screens that need not be immediate, the project supports the provider’s asynchronous batch mode, which processes requests at half price in exchange for a turnaround measured in minutes to hours rather than seconds. Because the screen is naturally run on a weekend before markets open, this trade-off is nearly free in practice and brings a full screen to roughly \$5. Batch mode is offered as a per-run choice rather than a default, preserving the option of immediate daytime screens at full price.

Two observations follow for anyone replicating the work. First, scraping or otherwise cheaply sourcing the enrichment data would save almost nothing, because enrichment is roughly five percent of cost; the expense lives in grounded agent reasoning. Second, the single most effective economy is asynchronous batch processing, followed by disciplined two-stage filtering, followed by retrieval depth—in that order.

6. Portfolio Construction and Management

The committee's output is converted into a real portfolio through a fixed, pre-committed process. A \$10,000 account is managed against the Dow Jones Industrial Average, the same universe the committee screens, so that the benchmark is the honest counterfactual: holding the index instead of the committee's selections.

6.1 From verdicts to positions

Only securities the committee rates Buy enter the portfolio; it is long-only by construction. Each position is sized in proportion to its committee conviction score, normalized so the selected names sum to the full deployment. A position the committee backs strongly receives more capital than a marginal Buy. The system renders this allocation as a portfolio chart with exact dollar amounts for the account size, which becomes the order list.

6.2 Execution

The weekly cycle is fixed. The screen is run on a weekend with markets closed, so that enrichment reflects a stable end-of-week snapshot and the committee's verdicts are available well before the next session. Orders are placed the following Monday as dollar-denominated market orders timed to 10:00 a.m. Eastern, fifteen to thirty minutes after the open, when spreads have tightened from their most volatile opening state. Dollar-denominated orders allow fractional-share execution and let the conviction-weighted allocation be entered directly without manual share-count conversion.

6.3 Weekly rebalancing

After the first week, the portfolio is rebalanced rather than rebuilt. Each weekend's fresh screen produces new target weights, which are reconciled against current holdings under three pre-committed rules:

- **Exit on removal.** Any holding the committee no longer rates Buy is sold in full.
- **Trade only meaningful drift.** An existing position is trimmed or topped up only if its weight has moved at least three percentage points from target; smaller drifts are left alone to avoid trading on noise.
- **Reinvest proceeds.** Cash freed by exits is redeployed into the new targets, prioritizing the most under-weight names, so the portfolio stays fully invested.

The reconciliation is computed against current total portfolio value—holdings at market plus cash—so that target percentages translate into correct dollar trades as the portfolio's value changes. The process produces a weekly trade ticket listing only the securities that require action; in a typical week this is a handful of trades rather than a full re-entry. Edge cases are handled explicitly: when forced sales would leave cash that the drift band cannot redeploy, proceeds are deployed into under-weight targets regardless of the band; and when fresh buys would exceed available cash, all buys are scaled proportionally to fit, preserving their relative weights without overdrawing the account.

6.4 The trade log

Every position is recorded in a trade log that captures the distinction between intent and execution: the committee's recommended weight, the actual fill price obtained, and the current market price for live profit-and-loss. This record is the raw material for results. Because it stores both the recommendation and the realized fill, it will permit later analysis of execution slippage separately from the committee's selection quality.

7. Benchmarking and the Measurement Plan

The portfolio is compared to the Dow Jones Industrial Average because the committee selects exclusively from the index's thirty constituents. This makes the benchmark a true alternative: any divergence in return reflects the committee's selection and weighting decisions, not exposure to a different universe. Performance will be evaluated on total return relative to the index and on the consistency of the committee's decisions week to week.

Several characteristics will be examined as results accumulate: whether the committee's pronounced tilt toward quality consumer and platform businesses helps or hurts in a given regime; how much realized return is lost to execution slippage and rebalancing turnover; and whether the deliberation phase changes outcomes relative to the blind vote often enough to justify its cost. Each of these is a property of the documented process, measurable against the fixed methodology established here.

No performance results are reported in this paper. The portfolio is newly established, and reporting partial or anecdotal returns before a meaningful track record exists would undermine the transparency the project is built to provide. Results will be reported separately, against the unchanged process described above.

8. Limitations and Transparency Commitments

- **Simulation, not the individuals.** The agents are language-model personas constrained to primary-source text. They are not the real investors and do not represent those investors' current views or endorse any holding.
- **Composition bias.** The five frameworks lack a primary macro voice, predisposing the committee toward quality consumer and platform names and away from cyclical and rate-driven businesses. This tilt is a measurable property of the strategy, not a defect to be concealed.
- **Corpus copyright.** The libraries contain copyrighted material and cannot be redistributed. The project can share code and methodology so others may construct their own libraries from sources they are entitled to use, but not the corpora themselves.
- **Model and data drift.** Both the underlying language models and the enrichment data evolve over time, which may affect reproducibility; runs are therefore dated and the process versioned.
- **Single committee.** Results will reflect one particular five-member composition. They speak to this committee, not to simulated-committee investing in general.

- **Not financial advice.** Nothing in the project constitutes a recommendation. It is a transparent experiment in whether a documented process produces measurable results.

9. Conclusion

Furton Research has built a complete apparatus for a specific, transparent experiment: five primary-source-grounded simulated investors, deliberating as a committee over the Dow 30, managing a real \$10,000 portfolio under a fixed weekly process benchmarked against the index. This paper has documented that apparatus end to end—the libraries, the four-phase engine, the retrieval pipeline, the deliberation and two-stage filtering, and the portfolio construction and rebalancing rules—together with the engineering economics that govern its operation, including the retrieval–caching trade-off that proved central to running the system affordably.

What remains is the part that cannot be rushed: time in the market, recorded honestly against a benchmark, under a process fixed in advance. By publishing the method before the results, the project commits to evaluating whatever happens against an unchanged standard. The question of whether a simulated committee of experts can invest is now an empirical one, and the instrument to answer it is in place.

Methodology Revisions

This paper is amended by dated, append-only revisions rather than silent rewrites. Each entry records what changed, why it changed, and the first weekly cycle the change governs. The pre-revision text elsewhere in the paper is left as written.

Revision of 2026-08-17: Reference-price sourcing

As originally built (Section 3.1), the enrichment phase’s web search doubled as the source of each security’s reference price — the mark used to value held positions and to size the weekly rebalancing ticket. In live operation those searched quotes proved silently stale at an accelerating rate: 5 of 30 names on the 2026-07-20 screen, then 9, 12, and 16 of 30 on the three weekly screens that followed, with nothing in the pipeline checking quote age. At the extreme, the portfolio’s largest holding was marked from a quote six and a half weeks old and 36.8 percent below the actual close, and one name returned an identical frozen quote four consecutive weeks. A manual weekly gate kept sizing correct from 2026-07-20 through 2026-08-10: quote dates were bucketed against the last trading close, stale-dated quotes were overridden with independently verified closes, and every override was recorded in the weekly records under `price_overrides`. Screener rankings preserve the raw engine view throughout.

Effective with the 2026-08-17 cycle, reference prices come from the Tiingo end-of-day API as settled closes of the last completed session. Marks are raw closes, never dividend-adjusted; the request window ends the day before the run, because during market hours the newest row is an in-flight value rather than a settled close; and a hard freshness gate of four calendar days covers weekends and single holidays. A security that fails the fetch or the gate fails its screen loudly and is excluded from the day’s

archive, which blocks ticket sizing — there is no fallback to searched prices. The verified close is also injected into the enrichment brief as the authoritative current price, so committee members reason on accurate valuation rather than whatever a search surfaced. The enrichment model still reports the price its own search finds; that figure is archived alongside each record as a cross-check (enrich_reported_price) and is never used for sizing. Every archived record now carries its price_source.

One smaller correction lands with the same revision. Where marks previously mixed settled Friday closes with live Monday quotes depending on what the out-of-date search data returned, they are now uniformly the prior session's settled close, so a run is reproducible at any hour. The public site's analytics generator moved to the same feed for equity and ETF series; the Dow index daily series remains on its prior source, as index levels sit outside the end-of-day API, and weekly benchmark levels remain manually recorded as before. Validation on 2026-08-10: the API's settled closes for all thirty index names matched independently hand-gathered closes exactly, and site analytics regenerated from the new feed matched the prior source on all 165 overlapping historical data points.

Primary Sources

The committee's five simulated investors are grounded exclusively in the primary-source corpora listed below. This list is current as of publication and is subject to expansion as additional primary material is collected. *These works are cited as the documented basis for each simulated investor; the project does not reproduce or redistribute their text.*

Warren Buffett

- Berkshire Hathaway, Inc. Shareholder Letters, 1977–2023. Published annually by Berkshire Hathaway and publicly available at [berkshirehathaway.com](https://www.berkshirehathaway.com).

Howard Marks

- Oaktree Capital Management client memos and published commentary by Howard Marks, drawn from the memo archive at oaktreecapital.com.

Joel Greenblatt

- You Can Be a Stock Market Genius. New York: Simon & Schuster, 1997.
- The Little Book That Beats the Market. Hoboken, NJ: John Wiley & Sons, 2005.
- The Little Book That Still Beats the Market. Hoboken, NJ: John Wiley & Sons, 2010.

The Big Secret for the Small Investor (2011) was reviewed and intentionally excluded; it advances value-weighted indexing as a retail-accessible alternative and introduces no new investment principles beyond those documented in the three included works.

Cathie Wood

- ARK Investment Management LLC. "Big Ideas" annual research reports: 2017, 2020, 2021, 2024, 2025, and 2026 editions.
- Interview: Chris Hayes, "Why Is This Happening?" (MSNBC podcast), conversation with Cathie Wood, March 2022.

- Interview: AI & Faith Town Hall, conversation with Cathie Wood, 2022.

A hybrid corpus was used because ARK's flagship reports are graphically dense and text-thin; interview transcripts were added to capture Wood's reasoning in prose.

Leopold Aschenbrenner

- Situational Awareness: The Decade Ahead. June 2024.
- Long-form interview with Dwarkesh Patel (podcast transcript), 2024.

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